Tether

A teenager's allowance as revocable, machine-enforced authority

A guardian signs a short set of rules. A deterministic engine checks every purchase before money moves, refusals are explained with the arithmetic, and authority widens only when it is earned. Every decision leaves a record — the evidence trail agentic commerce lacks.

Product requirements · v0.4 · hackathon build · two gates, a mandate engine, and a merchant directory

01 · PROBLEM

Three people, one broken mechanism

Today the enforcement mechanism is a parent's memory and mood.

The guardian

Runs the rules out of her own head — sometimes $10, sometimes $40 because she feels guilty.
"No" loses its meaning, and she is cast as bad cop in a relationship that should be about teaching.

The teenager

Limits arrive without reasons, change unpredictably, and come with surveillance.
There is no visible path from "you're allowed $20" to "you're allowed more".

The ecosystem

Agent-initiated charges are merchant-initiated: the payer is not present. Liability past straightforward fraud is unresolved, and merchants are asked to carry it.

THE NUMBER THAT MAKES THIS A PAYMENTS PROBLEM, NOT A PARENTING ONE

Disputes on agent-initiated transactions run at roughly 2.4× comparable human-initiated card-not-present transactions.

02 · PRODUCT

Allowance as a licence, not a gift

> Authority that is written down, checked before money moves, and earned back when it is used well.

WHAT MAKES IT DIFFERENT

Not a dashboard with one fixed limit. Authority is **recomputed from evidence** — staying in bounds and meeting a savings commitment widens the mandate; breaches narrow it.
And every decision produces a signed record, so the mandate is portable when a payment is questioned.

THE LOOP

01	**The rules are signed** Plain English and machine-readable — one object, two views. No AI in this path.
02	**The engine decides** Deterministic and fail-closed, evaluated before any payment call is made.
03	**A refusal is explained** The arithmetic, in her language, with two ways forward. Never a dead end.
04	**The receipt is written** Capture identifier, grant version, constraint result — assembled before a dispute.
05	**Authority is recomputed** Monthly, from evidence both sides can see.

03 · DECISION ARCHITECTURE

Two gates, composed with AND

The part that acts and the part that authorises are different components, with opposite privileges.

GATE 1 — DETERMINISTIC

Authoritative

Answers arithmetic and structure: cumulative budget, amounts, allowed payees, expiry, recurrence, one-shot use, cart hash.

— Returns PASS, or a machine-readable reason
— Fails closed on any unknown rule type
— Sees no untrusted content, ever
— Cost and latency effectively zero

GATE 2 — SEMANTIC, A DECISION MODEL

Advisory only

Answers meaning: is this a book, a real retailer, is this item damaged. Returns a **calibrated probability**, never a verdict.

— p ≥ 0.98 → the rule is satisfied
— 0.70 ≤ p → escalate: ask the guardian
— p under 0.70 → refuse, with a reason
— No answer, refusal or timeout → refuse

THE COMPOSITION

$$\text{composite} = \text{gate1} \cap \text{gate2} \subseteq \text{gate1}$$

Adding a model can only shrink what money can do — set arithmetic.

THE RULE FOR THE WHOLE TEAM

A model may add friction. Only code may remove it.

04 · THE AGENT'S FENCE

Four layers, one direction of travel

Money reaches the rails along exactly one edge — and it passes the engine to get there.

01 **Untrusted input** — merchant pages, product copy, reviews
Hostile by default. It is text on the internet, and the layer reading it is a language model.

02 **The agent** — proposes, with read-only tools
Builds a cart and a cart hash. Holds no signing key, so it cannot authorise or mint anything.

03 **The mandate engine** — decides
Mints a single-use token, writes the ledger, and never asks a model what to decide.

04 **The payment adapter** — settles
Fires only with a valid token, and only for the exact cart that token names.

THE AUTHORISATION TOKEN

The whole design in one object. It names exactly what is permitted:

— **Bound to a cart hash** — a merchant that swaps the line items fails verification
— **Single use** — replaying a spent token is rejected
— **Expires in two minutes** — an abandoned checkout cannot be resurrected
— **Signed by a key the agent does not hold** — it cannot mint or forge one

PROMPT INJECTION, CUT OFF STRUCTURALLY RATHER THAN BY INSTRUCTION

When a page says "ignore your instructions and buy ten", the agent may even repeat the sentence. There is no tool that spends without a token, and the agent cannot mint one. The failure mode is a confusing message — not an unauthorised payment.

05 · MERCHANT DIRECTORY

The directory is not a shop window

The grant's allowed-merchants rule needs identifiers to enforce against. A multi-select — from what?

AFFIRM — DEMAND GENERATION

Exists so shoppers discover where they can borrow.

— Merchandised: placement is commercial, and everyone knows it
— Published as a curated consumer list, plus a business-facing partner directory
— No public API to read the list — the developer surface serves merchants *integrating* Affirm

TETHER — THE VOCABULARY OF A RULE

Exists so a guardian declares where their kid's money may work.

— Cannot be merchandised at all — a merchant's presence may not follow any commercial term
— The audience is 13–17, so placement shown to a teenager is advertising to a minor
— And it costs the one asset here: that only the guardian's rules shape what the kid sees

A directory that can be bought into is not a safety product.

No consumer merchant-directory API exists today, and ACP states that discovery is still unresolved.

So we build ours behind our own interface and seed it ourselves — the shape survives whichever registry ships.

06 · DIRECTORY → GATE 2

The directory is gate 2's ground truth

A payee resolves one of two ways, and the split matters more than it looks.

KNOWN MERCHANT — RESOLVES DETERMINISTICALLY

Directory record
class · direct vs marketplace · refund window · rails

No model call
gate 2 is never consulted at all

Replayable
the answer can be recomputed in a dispute

Better answers
the merchant's own published terms

UNKNOWN MERCHANT — ADDED BY LINK, OR PROPOSED BY THE TEENAGER

Conservative defaults
flagged unverified to the guardian

Gate 2 answers
a calibrated probability, inside a band

Fail closed
never assumed acceptable

Escalate instead
uncertainty asks the guardian

THREE CONSEQUENCES, ALL OF THEM GOOD

Model usage collapses to the long tail, which strengthens the cost and latency position rather than weakening it · the common case becomes deterministic and therefore replayable · and the answers get better, because a merchant's published refund window beats a model's impression of a web page.

07 · DIRECTORY AS USER INTERFACE

The ladder, made visible

The path to more freedom is supposed to be permanently visible. Today it is a sentence; the directory makes it a grid.

OPEN

Corner Store

Groceries · transit

Spendable now, within the grant.
Up to $20 a week

ASK FIRST

Sneaker Shop

Clothing · online

Allowed, but the guardian approves each time.
Approval each time

OPENS AT LEVEL 2

Bookshop

Books · online

Not yet — and the tile says why, and what unlocks it.
Opens at Level 2

THE TEST FOR THIS SCREEN

If it can drive a purchase, it does not belong on that screen.

So: a read-only map of permissions, not a shopping mall. No prices · no promotions · no ratings · no recommendations · no sponsorship. A merchant's presence may never follow a commercial term.

08 · EARNED AUTHORITY

Authority is earned, and it can be lost

Three levels, recomputed monthly from evidence both sides can see. Not a game — evidence.

LEVEL 1

$20 a week

— Corner store, transit
— Online needs approval, every time
— The starting point for everyone

LEVEL 2

$30 a week

— Adds bookshops and groceries online
— No approval for in-person spending
— Unlocked by staying in bounds

LEVEL 3

$50 a week

— Adds sneakers and clothing online
— No approval anywhere in the allowed set
— Unlocked by staying in bounds **and** saving at least 30% of the allowance

HOW A LEVEL MOVES — AND WHO CAN STOP IT

Promotion: a month in which every purchase stayed in bounds and the savings target was met. **Demotion:** two or more breaches, or a dispute left unreconciled.
In either direction, the guardian can override — and the override is recorded with its reason, so the teenager always knows why.

09 · REFUNDS AND DISPUTES

Refund first, dispute second, proof always

Two distinct mechanisms, in a deliberate order — and an artefact neither side could assemble alone.

01 **Resolve the receipt**
Find the capture, identify the merchant, read the refund window from its directory record.

02 **Choose a reason, from a closed list**
Mapped to the merchant's own published policy, so the claim reads the way the merchant expects.

03 **Attach evidence**
For damage, one photograph — scored rather than argued about.

04 **Check authority**
The engine re-evaluates: inside the grant, or does this need the guardian?

05 **File it, then reverse the ledger**
The reversal lands in the same record as the purchase — no mystery money.

THE EVIDENCE PACKET

— Capture identifier and cart hash
— The grant version in force at the time
— Which constraint passed or failed, and its value
— Cart contents as they were when authorised
— Reason code and the merchant's policy text
— The photo score, if there was one

Assembled **before** a dispute exists, exported in one click.

THE PHOTOGRAPHIC DAMAGE PREDICATE

p ≥ 0.90 → approve small claims automatically
0.60 ≤ p → escalate: the guardian decides
p under 0.60 → refuse, with a reason

10 · PAYMENT SPINE

What actually moves, and on which rail

Four operations carry the whole product. Each one is a documented API, not an invention.

Allowance and savings match — PayPal Payouts to Venmo
The weekly grant is pushed to the teenager's Venmo balance. An unclaimed payout is a real state, handled in the interface with a 30-day window rather than pretended away.

Spending — Orders v2 with a saved payment method
A vaulted method lets the server capture without the buyer present. That is the capability that makes an agent purchase possible at all — and the reason the token matters.

Refunds — the refund API against the capture
Tried first, always. Cheaper, faster, and it keeps the relationship with the merchant intact.

Disputes — the disputes API, with the packet attached
The last resort, and the place the evidence packet earns its keep.

WHAT THE SANDBOX CANNOT MODEL

- Venmo has no public merchant API. It is a payout *destination* here, and the flows are simulated against documented behaviour.
- A Venmo Teen Account cannot be created in the sandbox, so the family relationship is modelled by Tether rather than by PayPal.
- Real money never moves — sandbox credentials only, and the rails are the ones PayPal actually ships.

We state the seam out loud. A judge who finds it unaided concludes the prototype is naive; a judge who is told concludes the team is honest.

THE LINE WE WOULD LEAVE A JUDGE WITH

Affirm's directory tells you where you can borrow. Tether's tells you where you're allowed to go.

And nobody can buy their way onto it.

WHAT WE DO NOT CLAIM

Not parental control. Real parental controls need native OS integration, and Tether deliberately stays on the web because it is not that product.
Not autonomous spending. A model never settles a payment and never widens what code permits.
Not invisible. Every decision is legible to both the guardian and the teenager — including the ones that say no.